

PCT/PCM3: Kernel Prediction Card

4. Histogram Interpretation

Image A has low contrast because most of its pixel intensities are concentrated in a narrow range around roughly intensity 100. Histogram equalisation would spread this concentrated mass over a much wider range of output intensities, making the image use more of the available intensity range.

5. CDF and Equalised Mapping

Given $L = 4$ and the normalised histogram $p = [0.50, 0.25, 0.125, 0.125]$:

CDF = $[0.50, 0.75, 0.875, 1.00]$.

Using $s_k = \text{round}(3 \times \text{CDF}(k))$,

the equalised mapping is: $s = [2, 2, 3, 3]$.

6. Kernel Prediction Card

K1 (all 1/9): Flat regions remain approximately unchanged because the kernel averages similar neighbouring values; edges are smoothed/blurred because different intensities are averaged together.

K2 (centre 8, ring -1): Flat regions give approximately zero response because equal neighbouring values cancel; edges produce a strong response because neighbouring intensities differ.

K3 (left +1, right -1 columns): Flat regions give approximately zero response; it detects directional intensity changes, particularly vertical edges.